SPE-226679-MS

Revolutionizing Fault-Stability Analysis: A Novel Approach to Managing Uncertainties in Unconventional Assets and Hydraulic Fracturing Stages Placement Optimization

E. Ibanez Martinez, J. Alvarellos Iglesias, A. Perez Carreras, E. López Puiggené, and J. Segura, Repsol SA, Technology Lab, Mostoles, Spain; R. C. Bezerra De Melo, Repsol SA, E&P Technical Division, Madrid, Spain

This paper was prepared for presentation at the SPE International Hydraulic Fracturing Technology Conference and Exhibition held in Muscat, Sultanate of Oman, 23 - 25 September 2025.

Abstract

The article presents a novel methodology for unconventional assets to determine fault stability while managing uncertainties. This approach is crucial for initial state, drilling conditions, and hydraulic fracture operations. It covers the initial stages at the field scale with high uncertainty in stresses and fault geometries to the PAD scale, where the main uncertainty lies in fault parameters.

Initially, stress state, rock properties, and fault geometry in an area have high uncertainties that are reduced during field development. An application software/tool was developed to evaluate fault stability under different scenarios of pressures. This tool identifies areas of greater risk of casing damage. At the PAD level, uncertainties in target geometry, stress state, rock properties, and fault geometry are reduced, but fault parameters (mainly friction angle) remain uncertain. A second tool was developed to determine the sliding probability of faults crossed in a well-pad.

The play has 3 distinctive pay zones/shale formations: upper, middle, and lower, confined by much higher strength formations above and below. The above formation, the upper-hard formation, is weak and highly stressed. The area of study did not have any data for a full anisotropic characterization; however, there was production data suggesting the connectivity in the staggered lateral wells. There is evidence of fault reactivation in the area due to leaks through the faults located in the weak and stressed formation where wells initiate the curvature, the heel section. Also, microseismic activity aligned with fault locations indicates some fault reactivation. The regional study was performed with the in-house developed methodology, locating high-risk areas in natural conditions, drilling pressures, and hydraulic fracturing (HF) conditions. At the pad level, a study example result is detailed in this paper. Stress-state was defined from pilot 1D Geomechanical Models, the formation tops surfaces, regional strains were applied, and rock properties were populated. Regarding fault properties and the impact on stability, the dip and the friction angle are the parameters with the most impact on stability. Montecarlo simulation with a range on friction angle was implemented in the example calibrated with an example of casing failure due to a fault reactivation in an HF operation. The results were used to change trajectories if the probability of sliding in the initial

state or drilling conditions when solving the issues or reducing/avoiding some stages due to possible fault reactivation if a tip of a fracture arrives to a fault with high risk in fracture closure pressures in the injection fluids at the tip.

Traditional workflows address uncertainties with independent variables. However, the field scale proposed methodology introduces a groundbreaking approach by managing uncertainties with all dependent variables, setting a new industry standard. Additionally, The application of the proposed methodology to the pad compared with a traditional code based on limit equilibrium method and only uncertainty in fault friction angle is presented, jointly with the proposed methodology theory. A precise 3D geomechanical model was performed to compare both methodologies, making it a powerful asset in the field.

Introduction

In the early phases of a project, it is crucial to know whether areas of an asset with fractures are in a critical state or on the edge of instability to make informed decisions. Unconventional assets often require less capital expenditure for new acreage exploration than traditional assets. Additionally, early development is characterized by significant uncertainty and low characterization.

There are several ways to assess fault stability, each with varying degrees of complexity. In wellbore fault stability analysis, parameter uncertainty has a considerable impact on oil drilling operations. The most common method is based on critical state theory. Finite Element Methods (FEM) and geomechanical modeling are widely used to predict failure around the wellbore, but their accuracy is heavily dependent on reliable input parameters such as rock strength, in-situ stresses, and pore pressures (Zoback, 2007; Fjaer et al., 2008). The mud weight window calculation, which determines safe drilling fluid densities to prevent both collapse and fracturing, becomes particularly sensitive to uncertainties in cohesion, friction angle, and stress magnitudes (Zhang, 2013). Probabilistic approaches, including Monte Carlo simulations, have gained prominence due to their ability to quantify the impact of parameter uncertainty on drilling safety (Moos et al., 2003), but the dependency of parameters is usually not fully faced. These methods transform deterministic geomechanical models into risk-based assessments, acknowledging that rock properties can vary significantly even within small intervals. The propagation of uncertainty through 1D and 3D geomechanical models often results in wide confidence intervals for critical parameters like collapse pressure and fracture gradient (Behrmann et al., 2000). Finally, the most sophisticated methods (that require a lot of information) are chemical-mechanical coupling models, essential for reactive shale formations, which are equally affected by uncertainties in clay content, fluid composition, and time-dependent properties (Van Oort, 2003). The inherent spatial variability of subsurface properties, combined with limited sampling data, makes uncertainty quantification crucial for safe and economic drilling operations, often requiring real-time model updates as new data becomes available during drilling. Finally, the uncertainties regarding the damage zone and the fault parameters required to model their behavior have, in all known cases, a high level of uncertainties and implications in all the methods described.

A method with low level of information and a full dependence of parameters is presented applicated to unconventional assets and compared against critical state method with a geomechanics approach.

Methodology

The methodology proposed follows the 1D Geomechanical Model stress calculation: elastic and strength rock properties, and effective stresses. The workflow has dependencies, and the uncertainties, for example in pore pressure, need be combined with minimum horizontal stress uncertainties. Once there is an effective stress definition surrounding the damage zones the proposed methodology can be applied.

Structural Causal Models Framework

The approach employs Structural Causal Models (SCMs), following Pearl's framework (2009). An SCM M is defined as a triplet (U, V, F), where U represents external (exogenous) variables determined by unknown mechanisms, V represents internal (endogenous) domain variables whose values are determined by other variables in the model, and F is a set of functions that assign values to each $V\_i \in V$ as $V\_i = f\_i(U\_i, Pa\_i)$, where Pa_i is a subset of variables in V (excluding V_i).

From M, we infer a directed acyclic graph G(M) representing conditional (in)dependencies between variables. The SCM defines joint probability distribution over domain variables V. In our implementation, domain variables (densities, tensions, etc.) are modeled by V, while U models uncertainty, with each $U\_i \in U$ following $U\_i \sim N(\mu\_i, \sigma\_i)$.

Safety Factor and Risk Assessment

The model ultimately computes a safety factor (Sf) that indicates operational safety. Given the non-parametric probability density function (PDF) of Sf, we calculate a fault stability factor (FSF):

$$FSF = \frac{1 - P_{10}(Sf)}{P_{90}(Sf) - P_{10}(Sf)} \tag{1}$$

where P_i refers to the i-th percentile. Alternatively, we compute the failure probability as the integral over the critical range:

$$P(0 < Sf < 1) = \int_0 1\, f\_Sf(sf)\, dsf$$

Variable Importance Analysis

Variable importance is assessed through two complementary approaches. First, we use marginal plots (M-plots) that visualize the safety factor as a function of a single variable of interest V_i, averaging over remaining variables V_{i}:

$$Sf = \hat{f}\_i(V\_i) = E\_V_{-i}[\hat{f}(V) | V\_i = v\_i] \tag{2}$$

To address potential spurious correlations from conditional expectations, we implement Accumulated Local Effects (ALE) plots. ALE plots divide the domain of V_i into intervals and calculate safety factor differences when intervening at interval endpoints, assuming local monotonicity. This approach provides causal interpretations and avoids generating improbable data.

Variable importance is quantified as the standard deviation of the resulting functions, producing tornado plots that rank variables by their influence on the safety factor.

Value of Information Framework

We propose a theoretical framework for assessing information value. Let FSF denote the a priori FSF value without observations, and FSF' represent the FSF when uncertainty in variable V_i is reduced from σ_i to σ'_i. The information gain G is calculated as:

$$G = \frac{Z}{V_i} \quad fi(\mu ij) \cdot (FSF - FSF_j) d\mu ij, \tag{3}$$

where FSF'_j is the resulting FSF when V_i follows a distribution with mean μ_ij and reduced standard deviation σ'_i. The mean of G indicates average risk change, while its standard deviation quantifies FSF variability following observation.

The final SMC developed and applicated to fault stability is presented in the Fig 1.

Figure 1—Workflow relation dependences used to compute uncertainties.

Following the methodology a Risk of Fault Reactivation (RFR) was defined as (4) derived from expression (1).

$$RFR = \frac{P[FS < 1.0] - 0.1}{0.9 - 0.1} \cdot 100 \tag{4}$$

Implementation Architecture

The system has been developed in Python and utilizes a model-view-controller architecture, with scientific libraries (NumPy, Pandas, SciPy.stats) for computation and Plotly for interactive visualization. The main components include developing models with a builder pattern, equation handling for domain-specific calculations, and comprehensive risk visualization graphing utilities such as Mohr circles and stress point graphics.

Real case, PAD level study

The methodology was applied and compared to critical state methods with a high level of stress definition in these three scenarios, using the tool created for unconventional reservoirs:

1. Initial (initial pressures and stresses before drilling.
2. During drilling to avoid drilling problems or destabilize faults crossed by the PAD wells.
3. During hydraulic fracturing work, the moment the "tip" (fracture front) reaches a fault.

The uncertainties focus on the stress state and horizontal properties in the 60 meters of reservoir, but above all the parameters of the faults that intersect (from their friction angle, geometry, cohesion, and others such as the initial opening).

The models have been generated from 1D geomechanical models in pilot wells that cover both the reservoir (mid and lower) and the formations above and below it, the stops and base of each of these formations, and a definition of the geometry of the faults where it has been possible to extract from the available seismic. From there, a "cake-layered" distribution of properties has been opted for in the absence of more information. Only in one area was a 3D model of initial stresses available.

As far as the faults are concerned, the tool allowed them to be studied plane by plane. So, each of the faults has been segmented into different segments using as a criterion a variation in azimuth or dip greater than 10 degrees.

As mentioned above, the tool developed for early stages of exploration for conventional CCUS storage and unconventional regional studies has been extended and applied to unconventional assets at PAD scale to 10 PADs. The main objective was to determine the capability of locating areas with high sliding risk, managing the geomechanics parameters uncertainties combined with the advantage that with a coarse definition of stresses, fault geometries, and rock properties (elasticity and strength), the results can be used at the same level to take the main decisions for the pilot: change trajectories or avoid some HF stages to reduce the fault reactivation risk.

For the pilot, a set of hypotheses have been defined: the overpressures are applied to the entire fault, not only to the point where the fracture is generated; the entire fault has the same angle of friction; and the same geomechanical model is used for both cases (from the 2.5D geomechanical model, the resulting stresses and pressures have been extracted and averaged for each formation).

The RFR parameter (Risk Fault Reactivation) has been used as the value is fault stability. In this case, since it was the first pilot in unconventional reservoirs, another analytical-probabilistic tool based on classical critical state failure stability (F-STABL) has been used.

Fault Friction Angle Calibration

In this case, a fault reactivation event occurs with a failure in one well of a PAD while a hydraulic fracture job was performed in a close previous well (HF in Fig. 2). A postmortem fine analysis was performed, determining a friction angle of the fault that makes only unstable the well that fails (sliding in Fig. 2). Sliding tendency (ST) and critical pore pressure coefficient (CPPC) were computed for the fault that was triangulated from fault sticks and densified since the maximum side of the triangles is less than 40 meters. Then the 2.5 geomechanics model was generated (as is detailed below) and used the 8 closer nodes to each triangle of the fault. After applying the limit equilibrium method, CPPC is compared with Shmin and ST, and the friction angle (FA) of the fault is increased since D4 well (hydraulic fracture injection) is not in a critical state and D6 is in a critical state.

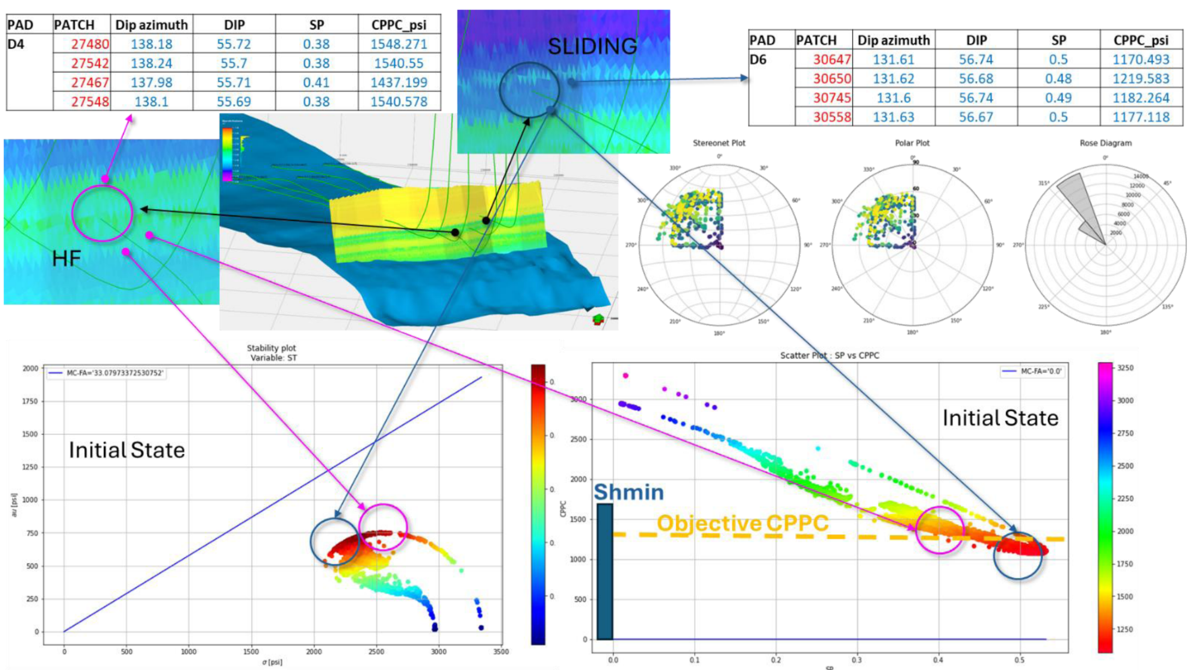


Figure 2—Retro-analysis with limit equilibrium methodology to determine the fault friction angle range.

The equilibrium was obtained over FA=38[deg] while the intact rock has a FA=45[deg]. These values are very high, and usually 20-25[deg] are the most common. But the hard data from the retro-analysis applicating the Limit equilibrium method is enough to constrain the values of friction angle of the faults from 38 to 45 [deg] for further analysis.

Rock and Stress Model Generation

For this exercise comparison, a 2.5 Geomechanical Model was generated. In the area there is a sort of 1D Geomechanical Model (1DGM) performed, all with vertical trajectories and with information in formations (1), (2), and (3). For any PAD, the surfaces of the formation's tops are available. Then, due to the lack of static models, the model is populated directly from the 1DGM. The faults have only available fault sticks.

In the Fig. 3. shows the PADs analysis as an example. In this case, 2 PADs are planned: one in the North (Well B) and one in the South (Well A). Two pilot 1DGM are available in the area with a good definition of stresses and rock properties. The zone with horizontal navigation is 60 meters, and then a mesh with 50x50 [m] cells and a vertical resolution of 2 [m] was defined to obtain an accurate enough stress and rock properties definition.

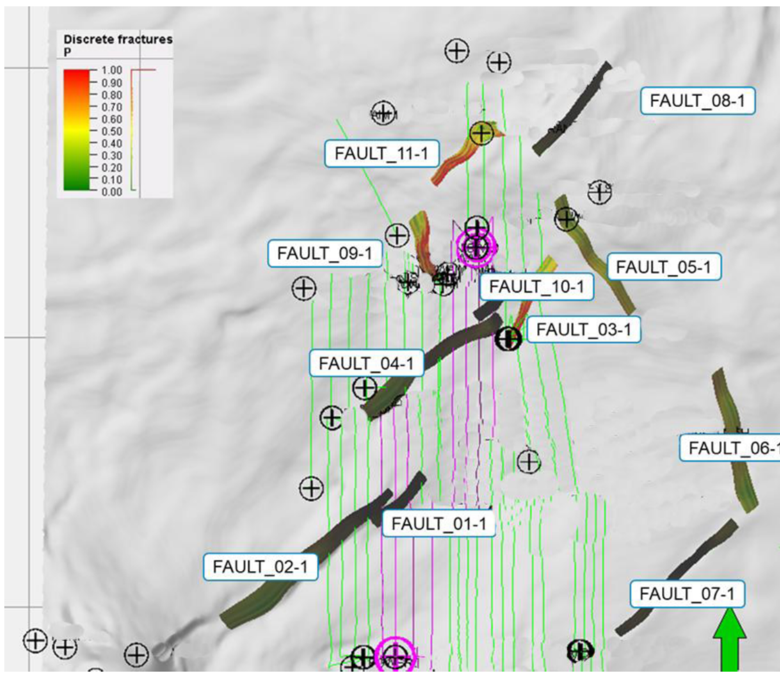


Figure 3—New Wells in purple and the faults to analyze at PAD scale (in green previous wells).

The population from the 1DGM was performed with two strategies: one for the rock properties and another for the stresses. The required rock properties (elastic properties) are density, Young's modulus, and Poisson's coefficient. An inverse distance approach was used to populate the following cells defined to keep as simple as possible, because there is no chance to assign mechanical lithologies first (see Fig 4). The vertical stress and pore pressure was interpolated based on depth.

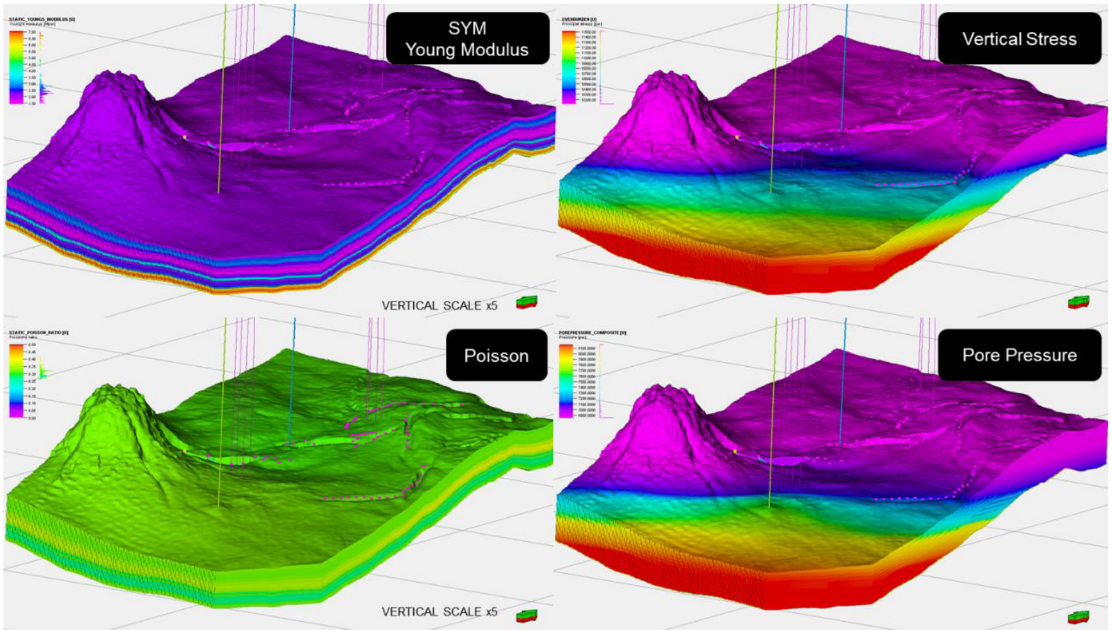


Figure 4—Model generated from formation tops and two 1D Geomechanical Model of Pilot wells with information over and under the objective formation.

To compute horizontal stresses a isotropic poro-elastic model was used using the previous knowledge of the area: horizontal tectonic strains, and azimuth of maximum horizontal stresses (see Fig 4 and Fig 5).

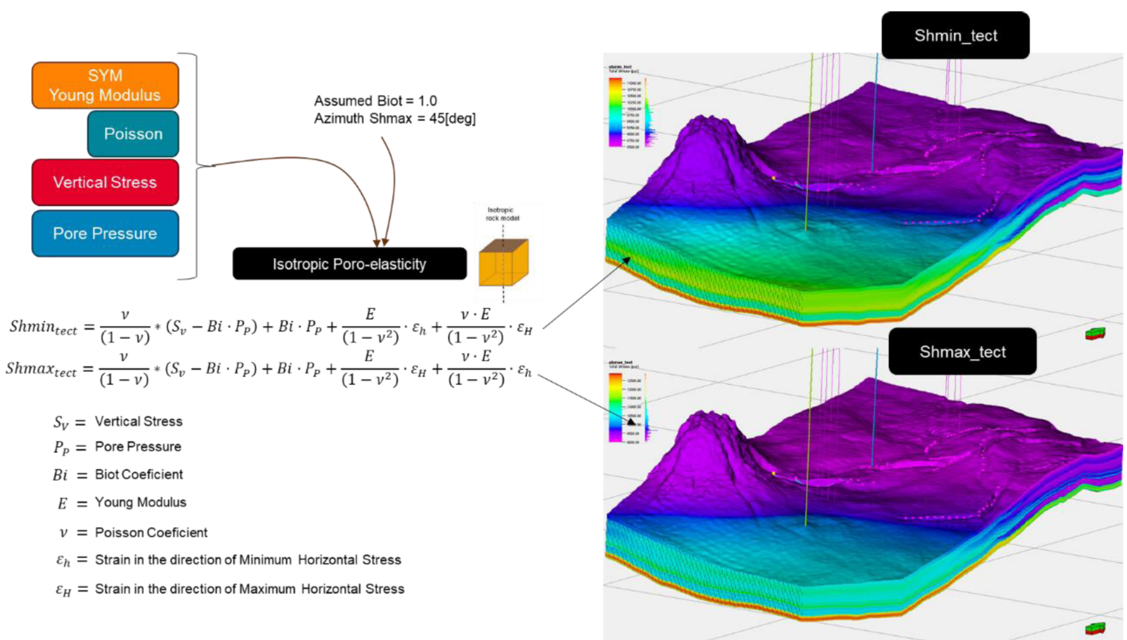
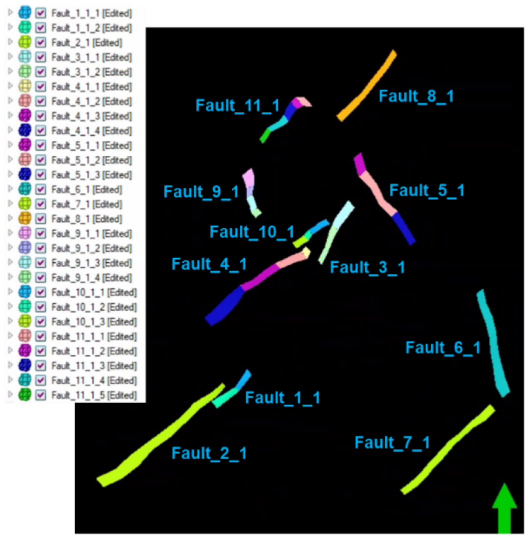


Figure 5—Horizontal Stress calculations using a isotropic poro-elastic model from vertical stresses and Pore pressures.

Fault Modeling

First, the available information (fault sticks) was meshed as a TIN with a flip strategy between each fault stick (interpretation of faults from high-resolution seismic in 2D sections). Then, over each center of the segments of each triangle, a triangle densification is performed while the maximum length is lower than 10 [m].

To apply the proposed methodology, a segmentation process is required, because each fault is computed as a quadrangular plane, given by azimuth and dip. The criterion followed was to segment whether there is a change in azimuth of more than 10 degrees or a dip of more than 10 degrees. For each segment, the mean and standard deviation of each surface have been calculated once reduced to a plane (Fig 6).



11 faults have been partitioned into 27 segments

Fault	dip_azimut_mean	dip_azimut_std	dip_angle_mean	dip_angle_std
fault_1_1_1	126.09	2.36	64.62	2.96
fault_1_1_2	145.58	3.12	61.4	4.88
fault_2_1	141.19	5.69	50.83	6.6
fault_3_1_1	301.36	5	53.61	2.12
fault_3_1_2	291.97	4.68	64.43	3.94
fault_4_1_1	126.25	9.08	58.4	2.92
fault_4_1_2	158.38	1.49	55.92	0.55
fault_4_1_3	148.18	5	56.9	1.11
fault_4_1_4	136.89	7.09	48.21	3.47
fault_5_1_1	76.22	0.51	50.86	0.54
fault_5_1_2	53.89	8.93	51.82	2.75
fault_5_1_3	61.24	1.44	54.52	1.11
fault_6_1	78.08	6.14	50.26	3.63
fault_7_1	136.87	5.83	57.2	2.97
fault_8_1	130.25	5.45	57.29	2.23
fault_9_1_1	253.59	7.85	53.53	2.82
fault_9_1_2	275.21	6.67	65.51	2.03
fault_9_1_3	250.24	5.15	63.06	1.08
fault_9_1_4	227.28	3.43	59.99	1.46
fault_10_1_1	149.81	6.07	64.84	1.55
fault_10_1_2	127.96	5.3	64.11	2.35
fault_10_1_3	146.39	3.86	62.85	1.74
fault_11_1_1_1	1.8	0.6	54.8	0.12
fault_11_1_1_2	359.2	0.2	54.8	0.12
fault_11_1_2	322.24	10.29	49.62	3.12
fault_11_1_3	299.73	7.55	56.22	2.5
fault_11_1_4	323.19	10.07	63.2	3.09
fault_11_1_5	303.19	3.43	66.06	1.02

Figure 6—Fault segmentation results from Fault Sticks.

For the limit equilibrium classical methodology to contrast the proposed methodology, the TIN is sufficient.

Results of Proposed methodology (USET)

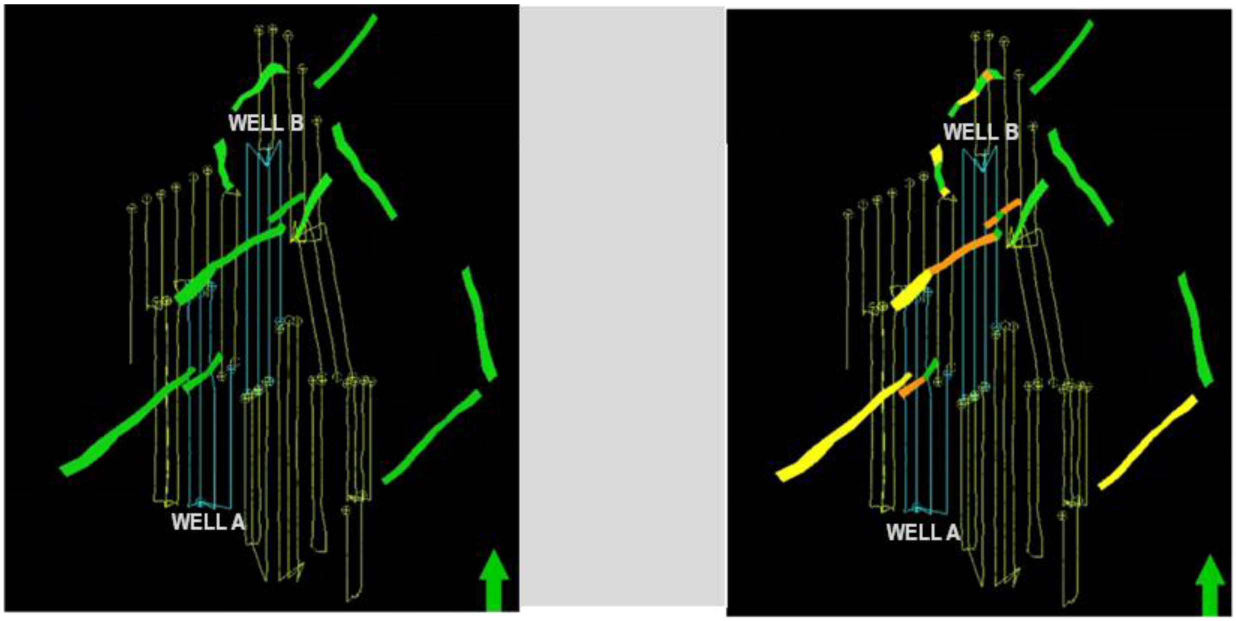
To implement the methodology, no spatial variations are allowed. For formations (1) and (2), statistical mean and sigma were calculated and implemented in the workflow to represent the values and uncertainties in stresses and rock properties. For each well, the navigated formation was utilized.

For cases (1) – Initial, (2) – Drilling conditions, and (3) – Case that HF tip reaches a fault, RFR was calculated from the uncertainties of all the inputs of the model with the following scenarios (see results in Fig 7):

1. Coefficients of friction angles of 0.6, 0.8 and 1.0 (calibrated with the back analysis of a previous case).
2. Scenarios: Initial conditions, drilling conditions (with an overpressure of 580[psi]) and hydraulic fracturing conditions (with a fluid overpressure of 1740[psi]).

To define the levels of risk the following criteria is used:

- If (RFR (cfa = 0.8) <0 AND RFR(cfa = 1.0)) = GREEN
- If (RFR (cfa = 0.8) <0 OR RFR(cfa = 1.0)) = YELLOW
- If (RFR (cfa = 1.0) <2 and RFR(cfa = 0.8)<20) = ORANGE
- If (RFR > 10) = RED



Fault Reactivation Risk for pp increment of 580 psi (4 MPa) over the in-situ pp. Representative of drilling conditions.

Fault Reactivation Risk for pp increment of 1740 psi (12 MPa) over the in-situ pp. Representative of HF conditions.

Figure 7—Result of the proposed methodology considering a coefficient of the friction angle of the faults between 0.8 and 1.0. In the left in drilling conditions and hydraulic fracture pressures at the right.

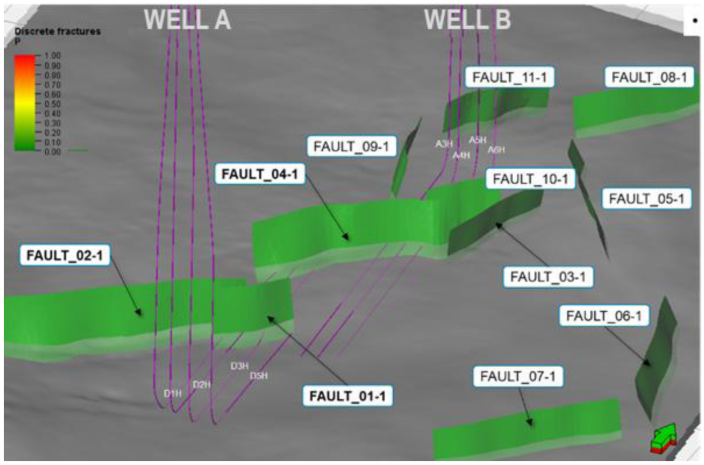
The results are presented in Fig. 7 for the initial case (1) and drilling conditions case (2) at the left and the case where a tip of a hydraulic fracture reaches any of these faults (3). For cases (1) and (2), all faults are in green (low reactivation risk), while in case (3), the PAD of WELLS A reaches some faults in yellow and orange risk, and the PAD of WELLS B reaches green (only one well of the PAD), yellow, and orange.

Then, the recommendations from the analysis are that the wells of each pad can be drilled with low risk of fault reactivation. But when the hydraulic fracture jobs are performed, the risk is high for arranging sections, and some actions are required. The orange areas are not in a critical state for all considered friction angles of the fault plane, but there is a high risk of fault reactivation if a tip of a fracture reaches the fault plane, and then previous wells that cross the fault can be lost (and their production) due to a slip (and failure of the casing) or proppant the fault and limit the production to the fault surrounding.

Depending on the environment of each PAD, there are some criteria to satisfy: production, neighbors, HSE, in this case the decision was to avoid the stages close to orange and red faults and use a safety distance to the fault.

Results of Limit Equilibrium Classic methodology (FSTABL)

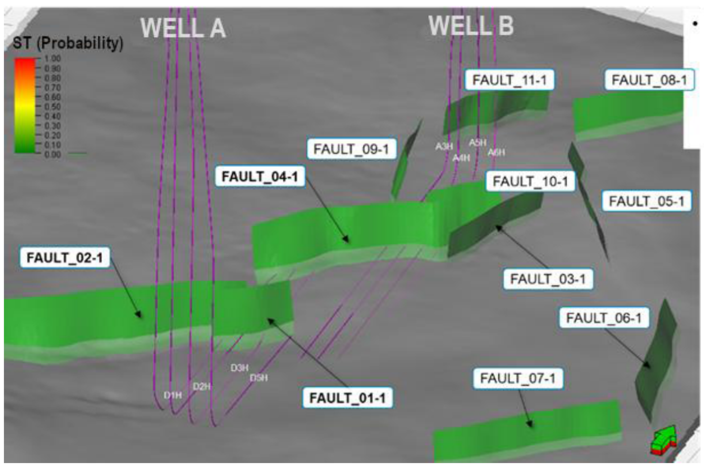
The classical approximation of limit equilibrium with uncertainty just in the friction angle was carried out as a verification calculation. The faults' TIN was used as surfaces. Values ranging from 0.6 to 1.0 have been taken into account in order to calculate the likelihood (see to Fig. 8—initial circumstances (1), Fig. 9—drilling conditions (2), and Fig. 9—injection conditions for hydraulic fracturing (3)).



- **A. Analysis Initial Conditions.**

Fault	Sliding Probability
Fault_01-1	0%
Fault_02-1	0%
Fault_04-1	0%
Fault_10-1	0%

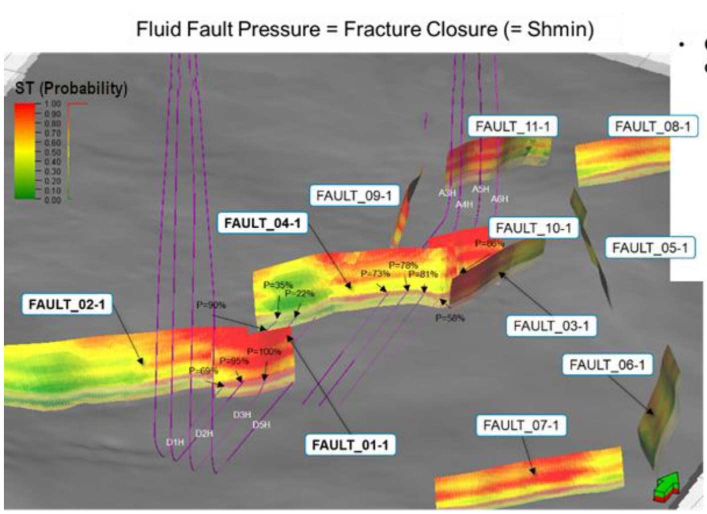
Figure 8—Result of the Proposed methodology application for the initial state. With the same results for the Limit equilibrium methodology with only uncertainty in the friction angle of the faults.



- **B. Analysis in Drilling conditions.**
  - Assumed 580[psi] over the Pore Pressure.
  - In Drilling conditions all faults have low sliding risk (green).

Fault	Sliding Probability
Fault_01-1	0%
Fault_02-1	0%
Fault_04-1	0%
Fault_10-1	0%

Figure 9—Sliding tendency probability in drilling conditions using Limit equilibrium analysis.



- **C. Analysis in Hydraulic Fracture conditions**
  - Probabilities in the table
  - Intersection points have higher sliding probability:
    - WELL A : F01=70-100%, F2 =90%, F04=22-35%
    - WELL B : F04=73-81%, F10=86%

Fault	Sliding Probability
Fault_01-1	26%
Fault_02-1	19%
Fault_04-1	37%
Fault_10-1	50%

Figure 10—Result with the classical limit equilibrium approximation for the analysis in the case of a hydraulic fracture reaching a fault

In this processing with FSTABL (also developed in-house in our R&D center), the meshes are polished to the resolution of the 2.5D model (10[m]) and interpolated in tensor on each triangle element of the fault with the 8 elements nearest to the center. Then they are projected onto the plane, and the limit equilibrium approach is used to the perpendicular and tangential stresses, yielding stability as the quotient of destabilizing and stabilizing forces. Cohesion of the fault was assumed to be zero.

The initial conditions (1) and drilling conditions have a low risk of fault reactivation (P[ST]=0%), however the scenario (3) has values greater than 30%, which for this project is deemed a significant risk of fault reactivation. A 10% maximum is typically considered the limit, but in this case, utilizing the previously described real scenario, it was extended to 30% to match the real case with fault reactivation.

Method Comparison and Discussion

Two different methods have been presented and applied to a real case. The usual (FSTABL) requires a moderate level of information: fault geometry and a stress and rock properties model (at least 1D or 2.5D). The proposed methodology (USET) requires just a single value of stress, rock properties (with the uncertainties), and azimuth & dip for each fault plane considered.

The results, although the different nature of the indexes used (RFR and ST) are comparable for the PADs analyzed. For initial state and drilling conditions, the conclusion is the same: low risk.

For the case (3) both methods rely in high-risk zones of the planned PADs. Both results are presented in the Fig 11:

- Fault 01-1 is orange in USET analysis, while in the FSTABL is mainly red and yellow (> P[ST]=30%). Both methods rely on the recommendation to drill the wells but avoid the stages planned "near" to the fault.
- Fault 04-1 is yellow for PAD-A Wells with USET methodology, while with FSTABL it is <15% of sliding risk. That means that the risk is low to medium, and hydraulic fracture Jobs can be executed with high control of pumping curves to detect any leak through the fault. For PAD-B wells, the risk is high in both methods, and avoiding the HF stages was recommended for both methods.
- Fault 10-1. For both methods the risk is high (orange for USET and red-yellow for FSTABL), and then the recommendation was to avoid the stages "near" to the fault plane.

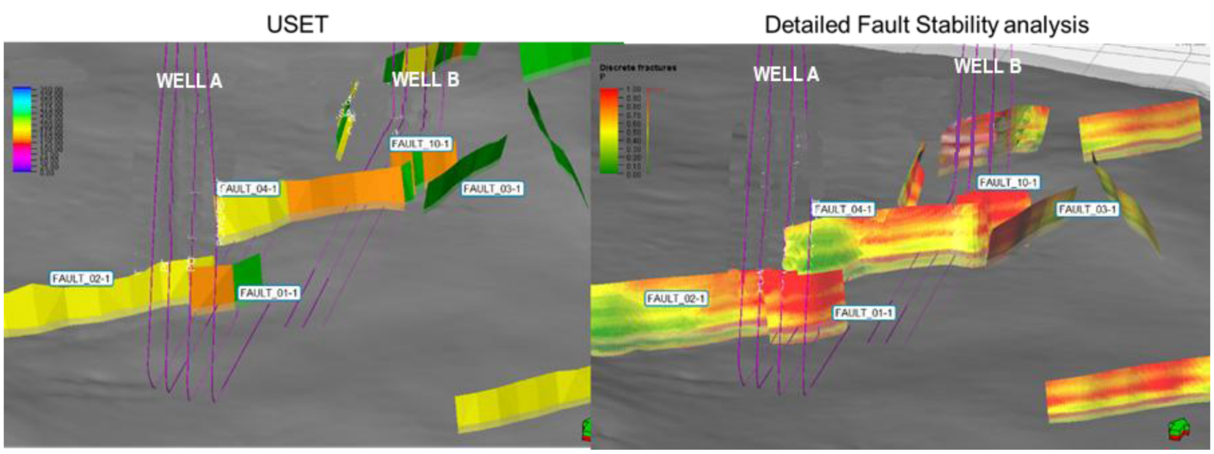


Figure 11—Comparison between both analyses for the case of hydraulic fracturing. At the left the results of USET with the risk defined with RFR parameter, and at the right the results of FSTABL for the same PAD with de Probability of Sliding tendency (P[ST]) as parameter of the risk.

The result of the study shows that the new methodology (USET) is applicable to this type of reservoir in a phase of determining potentially dangerous faults with low information and managing uncertainty in all its calculation parameters, since compared to the analytical tool (FSTABL), it is more accurate but requires more information.

Conclusion

The revised methodology, which takes into consideration all the uncertainties and dependencies, was presented. Although it was originally developed for conventional exploratory assets and CCUS evaluation in the initial stages, its adaptation to unconventional assets (where there are typically several earlier wells drilled) has been proved to be successful, both at the asset level for exploration of new accourages and at the PAD level.

A comparison with a more classic analytical tool was offered, with the only uncertainties being the friction angle of the faults investigated. The results for the ten PADs analyzed were successful.

The new methodology was able to determine the level of risk of a sort of fault with coarse definition and a low level of information for different scenarios (fluid overpressure ranges). Manages the uncertainties of each parameter with the dependencies, and if any parameter reduces the uncertainties (for example, if some pore pressure measures appear), it can be included in the analysis, and through the dependencies, reevaluate the fault stability. For a redefinition of the horizontal trajectory of the wells of a PAD to accommodate the best locations to cross a fault is required the FSTABL approach and finite element modeling (FEM) are required to model the fault behavior with complex models (that can accumulate energy) with more parameters and physical phenomena to be more accurate about the risk of fault reactivation.

Acknowledgment

We would like to express our gratitude to the Society of Petroleum Engineers (SPE) for providing the opportunity to publish this paper in their prestigious conference. We would also like to acknowledge the invaluable contributions of our colleagues who assisted with the data collection, analysis, and interpretation, without which this study would not have been possible.

Furthermore, we would like to extend our appreciation to the management of Repsol for authorizing the publication of this paper, and for their continued support of our efforts. We also acknowledge the contributions of the reviewers and editors for their valuable feedback and suggestions, which have greatly improved the quality of this paper.

Finally, we extend our thanks to all those who have contributed to this study in ways that may not be mentioned here. Your support and encouragement have been essential to our success.

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